«requirement»

Unflyable clearance refused

id = HLR-CRW-02
text = The flight crew shall accept a clearance only if the aircraft can fly it, and shall answer 'unable' when it cannot.

«deriveReqt»

«requirement»

Clearance within aircraft capability

id = SLR-CLR-10
text = The NAS shall issue only clearances that the aircraft can fly.

«requirement»

Conflict resolved with flyable clearance

id = HLR-ATC-04
text = Air traffic control shall resolve each predicted conflict with a clearance that every aircraft it moves can fly.
also from SLR-CLR-01

«deriveReqt»

«deriveReqt»

«requirement» [improved]

Performance share provided

id = HLR-OPS-05
text = Flight operations shall make each flight's current aircraft state and performance, and the airline's priorities for it, available to the party that decides a change to that flight's trajectory.

«deriveReqt»

«requirement» [improved]

Performance share used in resolution

id = HLR-ATC-08
text = ATC decision support shall use the best information available to it about each flight, including what the flight's airline shares.

«deriveReqt»

«requirement» [improved]

Decision information to deciding party

id = SLR-INF-02
text = The NAS shall give the party that decides a change to an aircraft's trajectory the information needed to preserve safety and pursue optimality.
also from SN-AIR-05

«deriveReqt»

«requirement»

Aircraft state shared

id = HLR-ACF-05
text = The aircraft shall make its current state available for sharing with the party that decides changes to trajectories.

«deriveReqt»

«deriveReqt»

«requirement»

Flyable and timely instructions

id = SN-CRW-03
text = The NAS shall give the flight crew only trajectory clearances and instructions that the aircraft can fly, with enough time to set them up.
owner: StakeholderRequirements::FlightCrews (Flight crew)

«deriveReqt»

«requirement»

Onboard performance predictions

id = SLR-EXE-03
text = The NAS shall predict each flight's arrival time, fuel at destination, optimum altitude and maximum altitude from the aircraft's estimated performance.
also from SN-AIR-01

«deriveReqt»

«requirement»

Better trajectory requested

id = HLR-CRW-03
text = The flight crew shall ask air traffic control for a change of route, altitude or speed when the crew deems it beneficial, optionally in collaboration with the dispatcher.
also from SLR-CLR-09

«deriveReqt»

«deriveReqt»

«requirement»

Weights and cost index entered in FMS

id = HLR-CRW-05
text = The flight crew shall enter the load sheet weights and the released cost index into the flight management system before departure.
also from SLR-PLN-07

«deriveReqt»

«deriveReqt»

«requirement»

FMS predicts performance

id = HLR-ACF-01
text = The flight management system shall predict the flight's arrival time, fuel at the destination, optimum altitude and maximum altitude from the entered weights, the cost index and the forecast winds.

«deriveReqt»

«requirement»

Dispatcher told of trajectory changes

id = HLR-CRW-04
text = The flight crew shall tell the dispatcher of each change to the cleared trajectory.
also from SLR-INF-07, SLR-CLR-04

«deriveReqt»

«requirement»

Low fuel prediction alert

id = SLR-EXE-06
text = The NAS shall alert the flight crew, dispatchers and controllers when the fuel predicted at the destination falls below the required reserve.
also from SN-GOV-01, SN-PAX-01, SN-CRW-02

«deriveReqt»

«requirement»

Flight re planned with crew

id = HLR-OPS-06
text = The dispatcher shall re-plan the flight with its crew when a trajectory change puts the arrival time, the fuel at the destination or flight safety at risk.
also from SLR-CLR-04, SLR-INF-07

«deriveReqt»

«requirement»

Crew alerted to low predicted fuel

id = HLR-ACF-03
text = The flight management system shall alert the flight crew when the fuel it predicts at the destination falls below the required reserve.

Circle-plus: ownership (the package or requirement on the left owns those on the right). Dashed arrow: deriveReqt, pointing from the derived requirement to its source. Grey: out of study (D-031). Lilac: background. Orange border: improved architecture (D-018).
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